CardioFusion

ECG model progress

Single recordings, four-channel input, and five-second segments

Supervisor progress update

1. The first model used separate segments

Each site recording was split into five-second segments and treated as a separate input.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.6858	63.8%	22.0%	64.2%	32.8%
Test	0.7921	65.8%	8.8%	77.5%	15.9%

A simple CNN with one input channel.

It caught some HFrEF cases, but produced too many false alarms.

Test confusion matrix

Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	1802	958
HFrEF	27	93

Accuracy is overall. Precision, recall and F1 refer to HFrEF.
Historical evaluation: 2,400 validation segments and 2,880 test segments. Cutoff 0.5452.

2. Removing segmentation was not enough

I used the full 30 seconds for training and changed the loss to focal loss.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.6995	76.5%	30.3%	54.5%	39.0%
Legacy test*	0.6160	62.7%	6.5%	59.2%	11.7%

Validation improved a little.

The model still saw one site at a time.
It was missing the patient grouping.

Test confusion matrix

Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	1735	1025
HFrEF	49	71

Accuracy is overall. Precision, recall and F1 refer to HFrEF.
*Saved test run still used segments. Validation: 400 recordings. This was not a matched test setup.

3. Grouping the four channels made the big difference

One patient now gives one input: APEX, LLSB, LUSB and RUSB, each with 30 seconds.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.9564	91.0%	57.9%	91.7%	71.0%
Test	0.9983	96.7%	55.6%	100.0%	71.4%

The CNN now takes four channels together.
Deeper residual blocks learn richer features.

The model can combine information from the same patient across the four sites.

Test confusion matrix

Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	111	4
HFrEF	0	5

Accuracy is overall. Precision, recall and F1 refer to HFrEF.
100 validation and 120 test patients. Grouping, CNN and split changed together; their effects are not isolated.

4. I trained another model using five seconds

The four channels stay together. A fresh model learns from six windows per training patient.

Results	AUROC	Accuracy	Precision	Recall	F1
Validation	0.8845	87.0%	47.6%	83.3%	60.6%
Test	0.9200	83.3%	17.4%	80.0%	28.6%

These results use only the first five seconds.

The model still finds useful information, but false positives rise from 4 to 19 compared with the full 30-second model.

Test confusion matrix

Actual / predicted	Non-HFrEF	HFrEF
Non-HFrEF	96	19
HFrEF	1	4

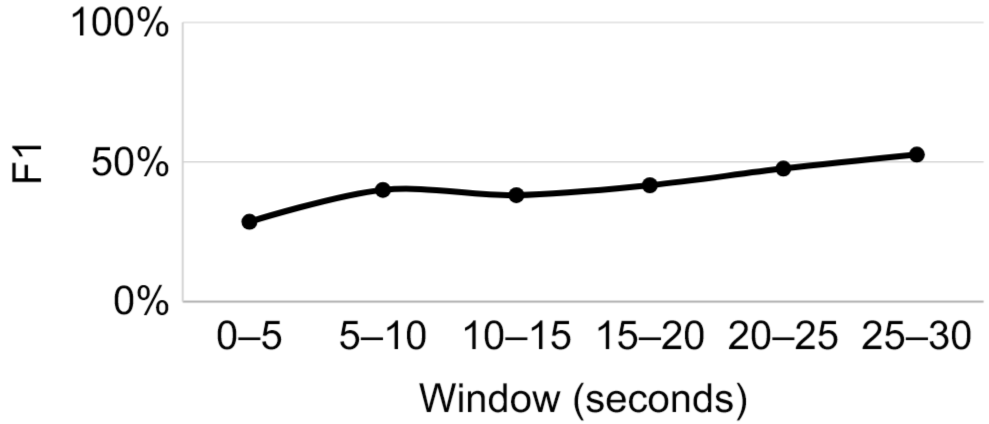
Accuracy is overall. Precision, recall and F1 refer to HFrEF.
Same patient split. Input is 4 × 2,500 samples. Five seconds means less signal required; speed was not measured.

5. Predictions change across segments

Test results for each five-second window, using the same model and cutoff.

All six predictions agree for 96 of 120 patients.

One HFrEF patient changes from negative in the first three windows to positive in the last three.

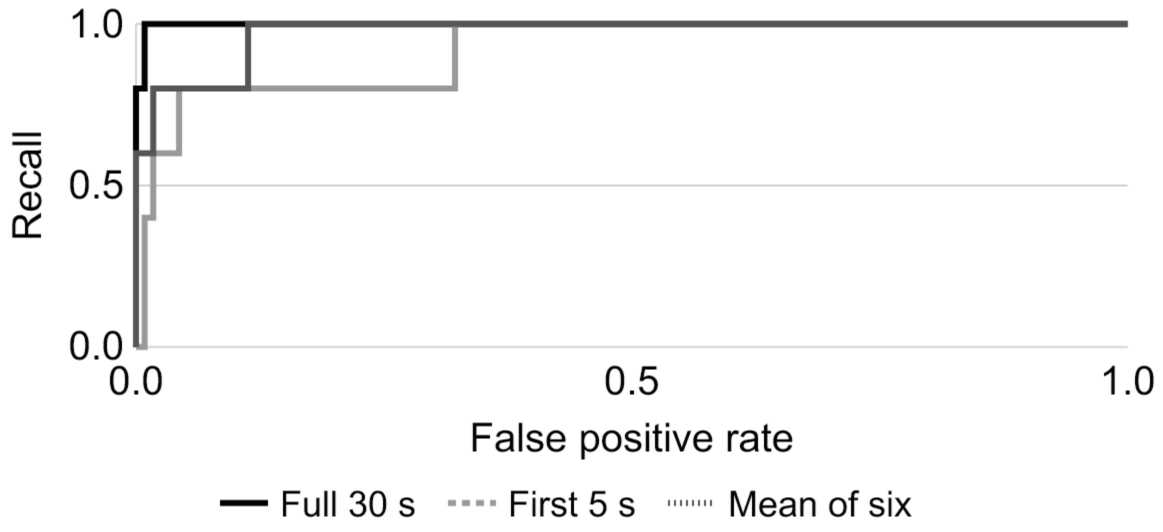


Results	AUROC	Accuracy	Precision	Recall	F1
0-5s	0.9200	83.3%	17.4%	80.0%	28.6%
5-10s	0.9357	90.0%	26.7%	80.0%	40.0%
10-15s	0.9670	89.2%	25.0%	80.0%	38.1%
15-20s	0.9635	88.3%	26.3%	100.0%	41.7%
20-25s	0.9739	90.8%	31.2%	100.0%	47.6%
25-30s	0.9896	92.5%	35.7%	100.0%	52.6%

120 test patients, including 5 HFrEF cases. Recall changes by 20 percentage points for one positive patient.
Repeated beats do not guarantee that every five-second window contains equally useful information.

6. Averaging segments helps

I compared one window, the mean of six window scores, and majority voting.



Averaging reduces false positives from 19 to 4.
It still misses one HFrEF case.

I would keep the full 30-second model for now and test across more patient folds.

Results	**AUROC**	**Accuracy**	**Precision**	**Recall**	**F1**
First 5 seconds	0.9200	83.3%	17.4%	80.0%	28.6%
Mean of six (30 s)	0.9739	95.8%	50.0%	80.0%	61.5%
Majority vote (30 s)	0.9661	92.5%	33.3%	80.0%	47.1%
Full 30-second model	0.9983	96.7%	55.6%	100.0%	71.4%

Test results for the same 120 patients. Averaging and voting still require 30 seconds of signal.
Thresholds were selected on validation. Mean scores break voting ties. Precision, recall and F1 are for HFrEF.